**Photosynthesis in Coral Symbionts under *Heat Stress* conditions!**

Did you know that algae live in symbiosis with coral, providing food for the coral in exchange for nutrients, carbon dioxide and shelter?

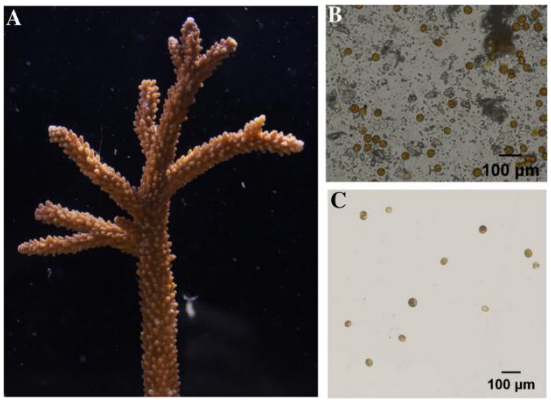
Despite their crucial role, corals are under increasing threat from climate change, which causes temperatures to increase, leading to coral bleaching and death. In this project you will analyse data on two coral-living algae species, *Durusdinium trenchii*, and *Cladocopium goreaui*. While *D. trenchii* is tolerant of high temperatures, *C. goreaui* is highly sensitive to such stress. Therefore, a comparative analysis could uncover what makes some corals better equipped to handle a warming world.

Using computational modelling, you will analyse data from ultrafast laser experiments on these two species. This analysis can uncover the dynamics of photosynthesis occurring within trillionths of a second. This can resolve the precise sequence of events from photon absorption to charge separation in the photosynthetic reaction center. **This project is perfect for students fascinated by photophysics, biophysics and energy processes in biological systems.**

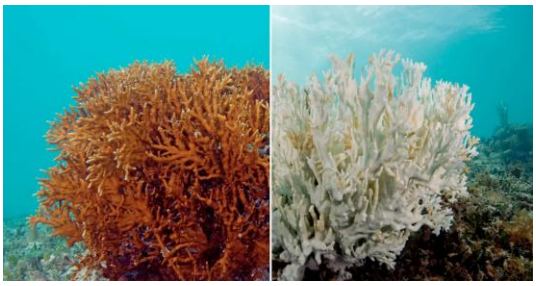
Join us in exploring the frontiers of life on our planet!

**Project supervisor**:
Dr Volha Chukhutsina, Assistant Professor in Biophysics of Photosynthesis group
v.u.chukhutsina@vu.nl

The coral Acropora Formosa (A) and the C. goreaui growing on it (B) and (C).




Healthy (left) and bleached (right) coral.



**Assignment title**
Programming and Testing of an Optical Spectroscopy Setup

**Assignment description**

Optical setups are key tools for investigating light-driven processes in physics, chemistry and material science. Transient absorption spectroscopy, in particular, uses ultrafast laser pulses to track how materials respond to light in real time. In our group, we are building and optimizing one of these setups. We are looking for students interested in programming and hands-on hardware.

Your main task will be:

- Programming the interaction between optical and electronic components (e.g., timing laser pulses and synchronizing high-resolution cameras) using LabVIEW.

Depending on your interests, additional tasks may include:

- Assembling hardware of optical components.
- Analyzing and visualizing data using Python.

You will gain practical experience in programming and hands-on mechanical work while testing a complex scientific instrument, integrating mechanical assembly, electronics, and software control.

**Primary supervisor name**
Prof. Dr. Volha Chukhutsina

**Primary supervisor email**
v.u.chukhutsina@vu.nl

**Department and research group**
Department of Physics and Astronomy - Biophysics of Photosynthesis and Energy group

**Where are you mainly located?**
De Boelelaan 1100, 1081 HZ Amsterdam, The Netherlands

**Where do you expect to have meetings with the student?**
Regular meetings at VU; online meetings possible in special circumstances..

**Location (supervisor at VU)**
Mainly at De Boelelaan 1100, 1081 HZ Amsterdam, in our research labs. Around 20% of tasks, such as code optimization, can be done remotely.

**How many students can be placed on this assignment?**
2

**Other supervisors names (optional)**
Dr. Giulia Lo Gerfo Morganti

**Other supervisors emails (optional)**
g.logerfomorganti@vu.nl

**Keywords (optional)**
Programming, automation, optical hardware, mechanics, lab work

**References or links (optional)**

[1] Berera, R., van Grondelle, R., & Kennis, J. T. (2009). Ultrafast transient absorption spectroscopy: principles and application to photosynthetic systems. Photosynthesis research, 101(2), 105-118. https://doi.org/10.1007/s11120-009-9454-y

**Requirements/student profile (optional)**

- Enjoy programming hardware (LabVIEW experience is a plus, but not required)
- Interested in mechanical and electrical integration of scientific instruments
- Optional: Curious about data analysis and Python scripting
- Optional: Motivated to work hands-on with optical and electronic components

**Comments for the organisers (optional)**
N/A